

Financial Time Series

T/4 Session #4

April 16, 2021

$$a_t = \varepsilon_t \sigma_t \quad \sigma_t = \sqrt{\omega + \alpha \varepsilon_{t-1}^2} \quad \text{ARCH(1)}$$

$\varepsilon_t \text{ iid } N(0,1)$

$$X_t = \phi_0 + \phi_1 X_{t-1} + \frac{\varepsilon_t}{a_t}$$

ARCH

→ GARCH

→ many variations
of GARCH to
Address some factor